**A DISSERTATION REPORT**

# EFFECT OF MACROECONOMIC VARIABLES ON SECTORAL STOCK RETURNS FOR THE PARTIAL FULFILLMENT OF THE DEGREE

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# CERTIFICATE

This is to certify that the dissertation entitled "Effect of Macroeconomic Variables on Sectoral Stock Returns: An Empirical Study of NSE Sectoral Indices" submitted by Shanu Singh (Enrollment No. 24010310253, SAP ID: 590010710),in partial fulfilment of the requirements for the award of the degree of Master of Business Administration (MBA Finance), School of Business, University of Petroleum and Energy Studies (UPES), Dehradun, is a bona fide record of original work carried out by the student under the guidance of Dr. Aniruddha Ghosh, Assistant Professor, School of Business, UPES.

The work embodied in this dissertation has not been submitted previously, either in full or in part, to any other university or institution for the award of any degree, diploma, or fellowship.

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# STUDENT DECLARATION

I hereby declare that the dissertation work entitled "Effect of Macroeconomic Variables on Sectoral Stock Returns: An Empirical Study of NSE Sectoral Indices" submitted to the School of Business, University of Petroleum and Energy Studies (UPES), is a record of original work done by me under the guidance of Dr. Aniruddha Ghosh, Assistant Professor, School of Business, UPES. The information and data given in this report are authentic to the best of my knowledge. This dissertation has not been submitted to any institution for the award of any degree, diploma, or fellowship, and is submitted purely for academic purposes under the MBA Finance programme.

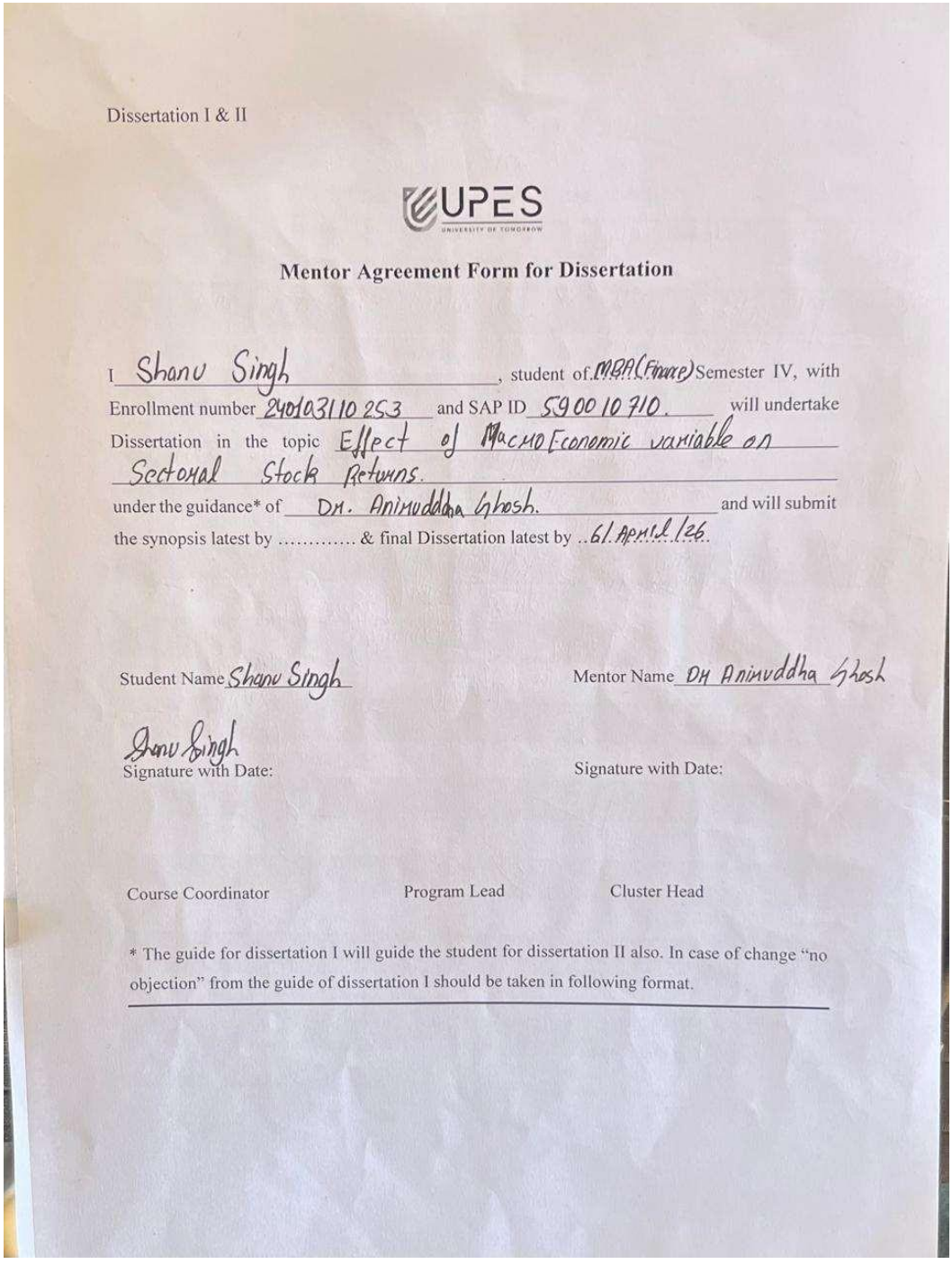
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# Mentor Agreement Form for Dissertation



Dissertation I & II

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**Mentor Agreement Form for Dissertation**

I Shanu Singh, student of MBA (Finance) Semester IV, with Enrollment number 24010311025 3 and SAP ID 590010710. will undertake Dissertation in the topic Effect of Macro Economic variable on Sectoral Stock Returns. under the guidance* of Dr. Aniruddha Ghosh. and will submit the synopsis latest by ..... & final Dissertation latest by ..6/ April /26.

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# ABSTRACT

This dissertation examines how selected macroeconomic variables affect sectoral stock returns, focusing on the Nifty Bank, Nifty IT, Nifty FMCG, and Nifty Auto indices. The empirical analysis is based on monthly data from September 2011 to May 2023 and uses a final returnbased econometric sample of 120 observations. The baseline explanatory variables are exchange rate, Nifty50 return, inflation, 10-year government bond yield, crude oil return, and India VIX return. The study applies descriptive statistics, correlation analysis, Ordinary Least Squares (OLS) regression, quantile regression, and impulse response function (IRF) analysis. The results show that Nifty50 return is the most consistent and statistically significant determinant of sectoral returns across all four sectors. In addition, the Banking sector is negatively affected by inflation, while the FMCG sector is adversely affected by crude oil return. Quantile regression shows that these macroeconomic effects are state-dependent: inflation has a stronger adverse effect on Bank returns in lower and median return states, while crude oil has its strongest negative effect on FMCG returns in the lower part of the distribution. IRF analysis further indicates that most macroeconomic shocks are short-run in nature and gradually converge toward equilibrium, pointing to dynamic stability in the selected sectoral systems. Overall, the dissertation contributes to the sectoral finance literature by showing that broad market return dominates sectoral performance, while selected macroeconomic variables continue to exert meaningful sector-specific and state-dependent effects.



Keywords: sectoral stock returns, macroeconomic variables, Nifty Bank, Nifty IT, Nifty FMCG, Nifty Auto, OLS, quantile regression, impulse response function.

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# Chapter- 1 Introduction

## 1.1 Background of the Study

Stock markets play a central role in mobilizing savings, allocating capital, and signaling investor expectations about future business conditions. In an emerging economy such as India, the link between macroeconomic conditions stock market performance is especially important because capital markets respond at the same time to domestic inflation, monetary conditions, exchangerate movements. commodity price shocks, and changes in investor sentiment.

Sectoral indices provide a more nuanced lens than aggregate market indices because industries differ in cost structure, demand sensitivity, regulatory exposure, and dependence on financial conditions. Banking is closely linked to inflation, credit conditions, and monetary policy; FMCG firms are often more sensitive to crude oil through transport, packaging, and input costs; Auto is cyclical and finance-sensitive: and IT is partly shaped by broader market conditions and external demand. These differences imply that the same macroeconomic shock can affect sectors in different ways.

Against this background, the study examines how selected macroeconomic variables influence sectoral stock returns by focusing on four major NSE indices-Nifty Bank, Nifty IT, Nifty FMCG, and Nifty Auto—and by using a structured econometric design to evaluate average effects, statedependent effects, and dynamic responses to shocks.

## 1.2 Statement of the Problem

A large body of financial literature suggests that macroeconomic variables influence stock market performance. Yet the magnitude, sign, and persistence of these effects differ across sectors and market states. Much of the evidence has focused on broad market indices such as Nifty50 or Sensex. Comparatively fewer studies investigate sectoral returns using a unified framework that combines mean-effect estimation, quantile-based asymmetry, and dynamic shock analysis.

This creates both a practical and an academic gap. Investors and portfolio managers need to know whether the Banking sector is more inflation-sensitive than the IT sector, whether oil shocks matter more for FMCG than for Auto, and whether these relationships differ during weak and strong market conditions. The present dissertation addresses this gap through a sector-wise empirical framework that combines OLS, quantile regression, and IRF analysis.

## 1.3 Research Questions

- How do selected macroeconomic variables influence sectoral stock returns?
- Do the Bank, IT, FMCG, and Auto sectors respond differently to market-wide and macroeconomic factors?
- Are the effects of macroeconomic variables uniform across weak, median, and strong return conditions ?
- How do sectoral returns dynamically respond over time to macroeconomic shocks?

## 1.4 Objectives of the Study

- To examine the descriptive and distributional characteristics of sectoral stock returns and selected macroeconomic variables .
- To analyze the relationship between selected macroeconomic variables and sectoral returns.
- To estimate the average effect of macroeconomic variables on sectoral returns using OLS regression.
- To examine whether macroeconomic effects differ across return states using quantile regression.
- To study the dynamic response of sectoral returns to macroeconomic shocks through impulse response analysis.

## 1.5 Hypotheses of the Study

- H1: Exchange rate has a significant impact on sectoral stock returns
- H2: Nifty50 return has a significant positive impact on sectoral stock returns
- H3: Inflation has a significant impact on sectoral stock returns
- H4: Government bond yield has a significant impact on sectoral stock returns
- H5: Crude oil return has a significant impact on sectoral stock returns
- H6: India VIX return has a significant impact on sectoral stock returns

## 1.6 Scope of the Study

The study is limited to four NSE sectoral indices: Nifty Bank, Nifty IT, Nifty FMCG, and Nifty Auto. The final return based econometric sample covers monthly observations from September 2011 to May 2023. The baseline explanatory variables are exchange rate, inflation, 10-year

government bond yield, crude oil return, India VIX return, and Nifty50 return. Other macroeconomic indicators such as repo rate, GDP growth, industrial production, and foreign institutional flows are beyond the scope of the baseline models.

## 1.7 Significance of the Study

The study is significant for investors, portfolio managers, policymakers, and researchers. For investors and portfolio managers, it identifies which sectors are more vulnerable to inflation, commodity shocks, uncertainty, and broad market movement. For policymakers, it offers evidence on how macroeconomic conditions are transmitted to sectoral equity performance. For academic research it contributes by integrating average-effect, distribution-sensitive, and dynamic-response models within one empirical framework.

## 1.8 Chapter Scheme

The dissertation is organized into five chapters. Chapter 1, introduces the background, objectives, hypotheses, and scope of the study. Chapter 2, reviews the relevant literature and develops the research gap. Chapter 3, presents the research methodology, data, variable definitions, and econometric framework. Chapter 4 reports the empirical analysis and interpretation of results. Chapter 5, summarizes the findings, conclusion, recommendations, implications, limitations, and scope for future research.

# CHAPTER 2: REVIEW OF LITERATURE

## 2.1 Introduction

This chapter reviews the theoretical and empirical literature on macroeconomic effects on stock returns, with special reference to India and sectoral stock indices. The Review is organized around theoretical foundations, international evidence, Indian evidence, sector-focused studies, and methodological literature related to OLS, quantile regression, and IRF analysis.

## 2.2 Theoretical Foundation

The principal theoretical basis of this dissertation is the Arbitrage Pricing Theory (APT), which argues that asset returns are driven by multiple systematic factors rather than by a single market factor alone. In the present study, this framework is operationalized by treating Nifty50 return, inflation, exchange rate, government bond yield, crude oil return, and India VIX return as systematic macro-financial factors that may influence sectoral stock returns.

Each factor fits the APT logic in a distinct way. Nifty50 return represents the broad market factor; inflation reflects domestic price pressure and the erosion of real returns; exchange rate captures external-sector conditions; government bond yield proxies the long-term financial environment; crude oil return represents a commodity cost shock; and India VIX return captures uncertainty and market fear

## 2.3 Review of Selected Studies

Foundational research by Chen, Roll, and Ross 1986 established that stock returns are linked to systematic economic forces, providing a durable theoretical basis for macro-factor asset-pricing studies. In the Indian context, Bhattacharya and Das 2014 and Agarwalla and Tuteja (2008) show that macroeconomic variables explain meaningful variation in Indian stock-market behavior.

More recent sectoral studies are especially relevant. Pandey 2022 and Chauhan at all 2025 indicate that NSE sectoral indices do not respond uniformly to macroeconomic variables. Gupta and Kumar (2023) and Maji, Laha, and Sur (2020) reinforce the idea that sectoral responses differ in the short and long run, while Tomar and Kesharwani 2022 show that monetary and price shocks can affect sectors asymmetrically.

The methodological literature provides equally strong support for the design of the present study. Koenker and Bassett 1978 and Koenker and Hallock 2001 demonstrate that quantile regression is valuable when the influence of explanatory variables differs across the conditional distribution of returns. Gjerde and Sættem 1999 provide a strong precedent for VAR/IRF methods to analyze the dynamic transmission of macroeconomic shocks through stock markets.

## 2.4 Synthesis of Literature

A synthesis of the literature yields four broad conclusions. First, macroeconomic variables matter for stock returns, but the direction and strength of their effects depend on economic context and sector structure. Second, sectoral heterogeneity is real and important. Third, mean-effect models alone may be insufficient because the impact of macroeconomic variables can differ across weak, median, and strong return states. Fourth, dynamic methods matter because shocks can affect returns over several periods rather than only contemporaneously.

## 2.5 Research Gap

Despite the relevance of prior literature, a clear empirical gap remains Many studies focus on the aggregate market rather than on sectoral indices. Studies that examine sectors often rely on only one econometric method without integrating average effects, distributional asymmetry, and dynamic responses in a single framework.

The present dissertation addresses this gap by examining four NSE sectoral indices using descriptive statistics, correlation analysis OLS regression, quantile regression, and IRF analysis within one integrated design. This allows the study to identify not only whether macroeconomic variables matter, but how they matter on average, when they matter across the return distribution, and how long their effects persist over time.

## 2.6 Conceptual Framework

The conceptual framework assumes that sectoral stock returns are jointly shaped by a broad market factor and by selected macroeconomic conditions. Nifty50 return captures overall market performance. Inflation captures domestic price pressure. Exchange rate reflects external-sector conditions. Government bond yield proxies long-term financial conditions. Crude oil return captures commodity cost shocks. India VIX return reflects uncertainty

The strength of these channels is expected to differ across sectors. Banking should be more inflation- and policy-sensitive, FMCG should be more exposed to crude oil related cost shocks Auto should be highly market sensitive and cyclical, and IT should remain partly market-driven but less directly explained by the domestic macro variables included in the baseline model.

## 2.7 Chapter Summary

This chapter has shown that the literature strongly supports a macro-sectoral approach to stockreturn analysis. It has also shown that a three-method design-OLS, quantile regression, and IRFis theoretically and methodologically justified. The next chapter presents the research methodology, data construction, variable selection logic, and model specification.

# CHAPTER 3: RESEARCH METHODOLOGY

## 3.1 Introduction

This chapter explains the research design, data structure, variable construction, and econometric methods used in the study. The methodology is intended to examine sectoral stock returns from three complementary perspectives: average effects, distribution-sensitive effects, and dynamic shock responses.

## 3.2 Research Design

The study adopts a quantitative research design. Sectoral stock returns are treated as dependent variables while selected macroeconomic and market related indicators are treated as explanatory variables. The analysis is conducted separately for each sector to identify differentiated responses rather than imposing single relationship across all sectors.

## 3.3 Data Source and Sample Period

The empirical analysis is based on monthly data. The final return-based econometric sample covers the period from September 2011 to May 2023 yielding 120 cleaned observations for the main regression dynamic analysis. The dataset was prepared from a structured Excel workbook that contains separate price and return sheets.

## 3.4 Variables Used in the Study

The study uses four dependent variables—Bank_Return, IT_Return, FMCG_Return, and Auto_Return and six baseline explanatory variables-Exchange_Rate, Nifty50_Return, Inflation, Govt_Bond 10Y, Crude_Oil_Return, and IndiaVIX Return. Each variable represents a distinct macro financial transmission channel and was retained on the basis of theory and empirical relevance.

## 3.5 Data Preparation and Cleaning

The data were first organized into a price-level dataset and a separate return dataset. Monthly frequency was maintained throughout. Sectoral stock returns were computed from price levels using percentage change, which is standard in financial econometrics and helps avoid spurious association from trending price levels.

Additional variables such as MIBOR, T-Bill, CBLO rate, and volume change were available in the broader dataset. However all of these were not included together in the final baseline model because several rate-type variables exhibited meaningful inter correlation in the correlation analysis raising concerns about multicollinearity. The final specification was therefore kept parsimonious and interpretable.

### 3.6 Correlation Analysis and Multicollinearity Logic

Before estimating the regression models correlation analysis was used to examine the association between sectoral returns and explanatory variables and to identify possible multicollinearity among regressors. Variables that overlapped strongly with one another were not loaded simultaneously into the same baseline model.

### 3.7 Econometric Tools and Rationale

The empirical framework consists of descriptive statistics, correlation analysis, Ordinary Least Squares, OLS regression, quantile regression, and impulse response function IRF analysis. Descriptive statistics summarize the distribution of variables. Correlation analysis screens association and multicollinearity. (OLS)estimates average effects. Quantile regression captures state-dependent effects across the return distribution. IRF traces the time-path of sectoral adjustment after one-time shocks.

### 3.8 Why OLS, Quantile Regression, and IRF Were Used Together

The three main econometric tools are complementary. OLS identifies the average impact of macroeconomic variables. Quantile regression shows whether that impact changes across weaker and stronger return states. IRF shows the time path of adjustment after a macroeconomic shock. Together, these methods provide a more complete empirical picture than any single method alone.

### 3.9 Model Specification

The baseline OLS specification for each sector is: $SectorReturn_t = \beta0 + \beta1\ Exchange_Rate_t + \beta2\ Nifty50_Return_t + \beta3\ Inflation_t + \beta4\ Govt_Bond_10Y_t + \beta5\ Crude_Oil_Return_t + \beta6\ IndiaVIX_Return_t + \varepsilon_t$.

Separate equations are estimated for Bank_Return, IT Return, FMCG Return, and Auto Return. Quantile regression uses the same explanatory variables but estimates the coefficients at different

points of the return distribution. For IRF analysis, sector wise small systems are used so that the response of each sector remains economically interpretable.

## 3.10 Hypothesis Testing and Significance Criteria

The study evaluates statistical significance primarily through p values. A variable is treated as statistically significant at the 5 percent level when p < 05. However, the dissertation does not rely on p-values alone. Interpretation is based jointly on statistical significance, coefficient sign and magnitude, and sectoral economic plausibility.

## 3.11 Limitations of the Methodology

The baseline models include only a selected set of macroeconomic variables and therefore do not capture every possible sector-specific driver. The sample is monthly, which improves consistency but limits very high-frequency inference. The IRF analysis is conducted in sectorwise small systems in order to preserve clarity and stability rather than in one large multivariate system.

## 3.12 Chapter Summary

This chapter has described the research design, sample structure, variable definitions, data preparation process, and econometric framework of the study. The methodology proceeds from descriptive and diagnostic analysis to average-effect, distribution-sensitive, and dynamicresponse modeling.

Table 3.1

Variable Definitions

| Variable | Definition | Role | Unit |
|---|---|---|---|
| Bank_Return | Monthly return of Nifty Bank index | Dependent | % |
| IT_Return | Monthly return of Nifty IT index | Dependent | % |
| FMCG_Return | Monthly return of Nifty FMCG index | Dependent | % |
| Auto_Return | Monthly return of Nifty Auto index | Dependent | % |
| Nifty50_Return | Monthly return of Nifty 50 index | Key market factor / control | % |
| Exchange_Rate | USD/INR exchange-rate variable used in return models | Independent | Rate |
| Inflation | Monthly inflation rate | Independent | % |

| | | | |
|---|---|---|---|
| Govt_Bond_10Y | 10-year government bond yield | Independent | % |
| Crude_Oil_Return | Monthly crude oil return | Independent | % |
| IndiaVIX_Return | Monthly India VIX return | Independent | % |

Source: Author's compilation based on the final cleaned monthly dataset.

Table 3.2

Econometric Tools and Rationale

| Tool | Purpose | Reason for Use |
|---|---|---|
| Descriptive statistics | To summarize average return, volatility, and distribution before modeling. | Financial series are often volatile and non-normal; summary statistics provide the first diagnostic picture. |
| Correlation analysis | To examine association and identify possible multicollinearity. | Highly correlated regressors can distort coefficient estimates if used together. |
| OLS regression | To estimate the average impact of selected macroeconomic variables on sectoral returns. | OLS provides the baseline meaneffect relationship and is widely used in macro-stock research. |
| Quantile regression | To estimate effects in low-return, median-return, and high-return states. | Macro effects may differ across the return distribution; mean effects alone may hide such heterogeneity. |
| IRF analysis | To trace the time-path of sectoral return response to one-time shocks. | Dynamic response analysis reveals whether shocks are temporary or persistent. |

Source: Author's compilation based on the research design of the dissertation.

# CHAPTER 4: DATA ANALYSIS, RESULTS AND INTERPRETATION

## 4.1 Introduction

This chapter presents the empirical analysis of the study. It focuses on four NSE sectoral indices Nifty Bank, Nifty IT, Nifty FMCG, and Nifty Auto and evaluates how their monthly returns respond to selected market-wide and macroeconomic variables.

The chapter begins with descriptive statistics, followed by correlation analysis, sector wise OLS regression, quantile regression, and impulse response function (IRF) analysis. This sequence allows the study to move from preliminary diagnostics to average-effect, state-dependent, and dynamic-response evidence.

## 4.2 Descriptive Statistics

Descriptive statistics provide the first empirical view of the dataset. They summarize the central tendency, dispersion, and shape of the return series and selected macroeconomic variables.

Table 4.1

Full Descriptive Statistics — Returns and Selected Macro Variables

| Variable | Mean | Std Dev | Min | Max | Skewness | Kurtosis | Character |
|---|---|---|---|---|---|---|---|
| Bank_Return | 1.633 | 8.514 | -26.118 | 30.807 | 0.571 | 2.119 | Aggressive, high-beta, fat-tailed |
| IT_Return | 1.617 | 6.504 | -16.226 | 22.485 | 0.268 | 0.568 | Moderate, marketlinked |
| FMCG_Return | 1.424 | 4.768 | -9.583 | 16.136 | 0.544 | 0.241 | Defensive, low-volatility |
| Auto_Return | 1.361 | 6.897 | -14.624 | 22.676 | 0.235 | 0.751 | Cyclical, marketsensitive |
| Nifty50_Return | 1.231 | 5.26 | -11.979 | 20.093 | 0.545 | 1.109 | Broad market benchmark |
| Inflation | 0.461 | 0.609 | -1.01 | 2.14 | 0.095 | -0.116 | Stable, nearnormal |
| MIBOR | 6.702 | 1.742 | 3.36 | 11.99 | -0.136 | 0.046 | Interest-rate level |
| Govt_Bond_10Y | 7.364 | 0.844 | 5.81 | 8.96 | -0.048 | -0.947 | Long-term rate environment |

| Crude_Oil_Return | 0.28 | 11.219 | -49.98 | 39.81 | -0.132 | 4.456 | High return |
|---|---|---|---|---|---|---|---|
| Variable | Mean | Std Dev | Min | Max | Skewness | Kurtosis | Character |
| | | | | | | | volatility |
| IndiaVIX_Return | 1.25 | 19.888 | -46.584 | 68.84 | 0.706 | 0.66 | Extreme fear spikes |

Source: Author's computation based on the final cleaned monthly dataset.

The Banking sector recorded the highest average return and the highest standard deviation among the four sectors, confirming its aggressive and high beta character. The FMCG sector displayed the lowest volatility and therefore appears comparatively defensive. Auto remained cyclical and market-sensitive, while IT occupied a moderate middle position. Crude oil and India VIX were especially volatile among the macro financial variables, making them meaningful candidates for return modeling.

## 4.3 Correlation Analysis

Correlation analysis was conducted to examine the degree of association between sectoral returns and the selected macroeconomic variables and to identify possible multicollinearity among explanatory variables.

Table 4.2

Selected Correlation Coefficients

| Variable 1 | Variable 2 | Correlation |
|---|---|---|
| Bank_Return | Nifty50_Return | .909*** |
| IT_Return | Nifty50_Return | .467*** |
| FMCG_Return | Nifty50_Return | .694*** |
| Auto_Return | Nifty50_Return | .796*** |
| Bank_Return | IndiaVIX_Return | -.305*** |
| FMCG_Return | IndiaVIX_Return | -.355*** |
| Auto_Return | Bank_Return | .747*** |
| Exchange_Rate | Govt_Bond_10Y | .643*** |

Source: Author's computation based on the final cleaned monthly dataset.

Note: *** $p < .001$.

All four sectoral returns are positively related to Nifty50 return but the magnitude differs substantially across sectors. The strongest market linkage is observed for Bank Return, followed by Auto Return, FMCG Return, and IT Return. India VIX is negatively related to sectoral returns, indicating that rising uncertainty tends to weaken sector performance. The correlation matrix

also supports a parsimonious final specification because some rate type variables move together strongly enough to raise multicollinearity concerns if used simultaneously.

## 4.4 Ordinary Least Squares (OLS) Regression Analysis

OLS regression was used to estimate the average effect of selected macroeconomic variables on sectoral returns. Separate models were estimated for Bank Return, IT Return, FMCG Return, and Auto Return.

Table 4.3

Comparative OLS Summary by Sector

| Sector | $R^2$ | Adjusted $R^2$ | F | Model p | N |
|---|---|---|---|---|---|
| Bank | 0.841 | 0.833 | 99.676 | 0.0 | 120 |
| IT | 0.229 | 0.188 | 5.603 | 0.0 | 120 |
| FMCG | 0.519 | 0.493 | 20.304 | 0.0 | 120 |
| Auto | 0.647 | 0.629 | 34.574 | 0.0 | 120 |

Source: Author's computation based on the final cleaned monthly dataset.

Table 4.3a

Full OLS Regression Coefficients — coefficient (p-value)

| Variable | Bank | IT | FMCG | Auto |
|---|---|---|---|---|
| Intercept | -4.149 (0.165) | 2.479 (0.621) | -1.814 (0.532) | -4.975 (0.168) |
| Exchange_Rate | -21.964 (0.919) | 215.089 (0.554) | 149.765 (0.477) | 152.164 (0.560) |
| Nifty50_Return | 1.511 (0.000) | 0.539 (0.000) | 0.622 (0.000) | 0.996 (0.000) |
| Inflation | -1.335 (0.014) | 0.467 (0.607) | -0.078 (0.882) | 0.520 (0.424) |
| Govt_Bond_10Y | 0.657 (0.198) | -0.679 (0.428) | 0.040 (0.936) | 0.349 (0.569) |
| Crude_Oil_Return | 0.010 (0.742) | 0.037 (0.474) | -0.066 (0.026) | 0.046 (0.216) |
| IndiaVIX_Return | 0.022 (0.208) | -0.005 (0.863) | -0.026 (0.128) | -0.013 (0.529) |

Source: Author's computation based on the final cleaned monthly dataset.

Note: Values are reported as coefficient (p-value).

The OLS results indicate that Nifty-50 Return is the most consistent and statistically significant determinant of sectoral stock returns across all four sectors. For the Banking sector Nifty50 Return is positive and highly significant, while Inflation exerts a significant negative effect. For FMCG, Nifty50 Return remains significant and Crude Oil Return is negative and significant. IT and Auto are primarily market-driven in the baseline mean-effect model.

### 4.4.1 Banking Sector

The OLS model for the Banking sector is statistically significant $F = 99.676$, $p < .001$, with $R^2 = .841$ and adjusted $R^2 = .833$. Nifty50_Return has a positive and highly significant effect on

Bank_Return, $\beta = 1.511$, $p < .001$, indicating that Bank Return moves more than proportionately with the broad market. Inflation has a negative and statistically significant effect, $\beta = -1.335$, $p = .014$ suggesting that inflationary pressure adversely affects banking returns.

#### 4.4.2 IT Sector

The IT sector model is statistically significant overall $F = 5.603$, $p < .001$, but its explanatory power is much lower than that of the Banking sector ($R^2 = .229$; adjusted $R^2 = .188$). Nifty50 Return is the only statistically significant explanatory variable, $\beta = 0.539$, $p < .001$. This indicates that the IT sector is primarily market-driven within the current specification.

#### 4.4.3 FMCG Sector

The FMCG model is statistically significant $F = 20.304$, $p < .001$, with $R^2 = .519$ and adjusted $R^2 = .493$. Nifty50_Return has a positive and highly significant effect, $\beta = 0.622$, $p < .001$, while Crude_Oil_Return has a negative and significant effect, $\beta = -0.066$, $p = .026$. The negative crude-oil coefficient suggests a cost transmission channel that is sector specific and economically meaningful for FMCG.

#### 4.4.4 Auto Sector

The Auto-sector model is statistically significant, $F = 34.574$, $p < .001$, with $R^2 = .647$ and adjusted $R^2 = .629$. Nifty50 Return is positive and highly significant, $\beta = 0.996$, $p < .001$, whereas the remaining macroeconomic variables are not statistically significant in the baseline model. Thus, Auto appears strongly market-linked and primarily market driven in the averageeffect framework.

### 4.5 Quantile Regression Analysis

Quantile regression was used to examine whether the influence of macroeconomic variables differs across the return distribution. In the revised analysis, wider quantiles 05, .25, .50, .75, and .95 were estimated to strengthen interpretation across extreme downside, middle, and upside states.

Table 4.4

Key Wider Quantile Regression Results

| Sector | Key Variable | Q0.05 | Q0.25 | Q0.50 | Q0.75 | Q0.95 |
|---|---|---|---|---|---|---|
| Bank | Inflation | -2.738<br>(0.134) | -1.634<br>(0.025) | -1.312<br>(0.021) | -0.997<br>(0.180) | -0.688<br>(0.467) |
| FMCG | Crude_Oil_Return | -0.139<br>(0.002) | -0.089<br>(0.009) | -0.070<br>(0.054) | -0.063<br>(0.222) | -0.027<br>(0.703) |

| | | | | | | |
|---|---|---|---|---|---|---|
| IT | Nifty50_Return | 0.331 (0.028) | 0.497 (0.001) | 0.683 (0.000) | 0.647 (0.000) | 0.579 (0.000) |
| Auto | Nifty50_Return | 1.030 (0.000) | 1.055 (0.000) | 1.096 (0.000) | 1.125 (0.000) | 0.978 (0.000) |

Source: Author's computation based on the final cleaned monthly dataset.

Note: Values are reported as coefficient (p-value).

The wider quantile results reinforce the sector-specific story. For the Banking sector, Inflation is most relevant in the lower and median quantiles: β = -1.634 (p = .025) at Q25 and β = -1.312 (p = .021) at Q50, but the effect weakens at Q75 and Q95. This indicates that inflation is more damaging when Banking-sector returns are weak or normal than when they are unusually strong.

For FMCG, Crude_Oil_Return is strongest in the lower tail of the return distribution: β = -0.139 (p = .002) at Q05 and β = -0.089 (p = .009) at Q25, with the effect weakening toward the upper quantiles. This asymmetry supports the interpretation that oil-related cost shocks matter most when FMCG returns are already under pressure. For IT and Auto, the dominant and stable explanatory variable remains Nifty50_Return across the wider quantile range.

## 4.6 Impulse Response Function (IRF) Analysis

IRF analysis was conducted to examine the dynamic response of sectoral returns to one-time shocks. While OLS and quantile regression emphasize contemporaneous relationships, IRF traces the time path of adjustment over subsequent months.

Table 4.5

Key IRF Response Values — Selected Shocks

| Month | Bank: Inflation Shock | FMCG: Crude Oil Shock | IT: Inflation Shock | Auto: Inflation Shock |
|---|---|---|---|---|
| 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 1.0 | -1.383 | 0.051 | 1.257 | 0.822 |
| 2.0 | 0.641 | 0.061 | 1.975 | 2.184 |
| 3.0 | 0.57 | -0.014 | 0.411 | 0.78 |
| 4.0 | -0.112 | -0.031 | -0.084 | -0.43 |
| 5.0 | -0.193 | 0.002 | -0.185 | -0.357 |

Source: Author's computation based on sector-wise VAR estimation.

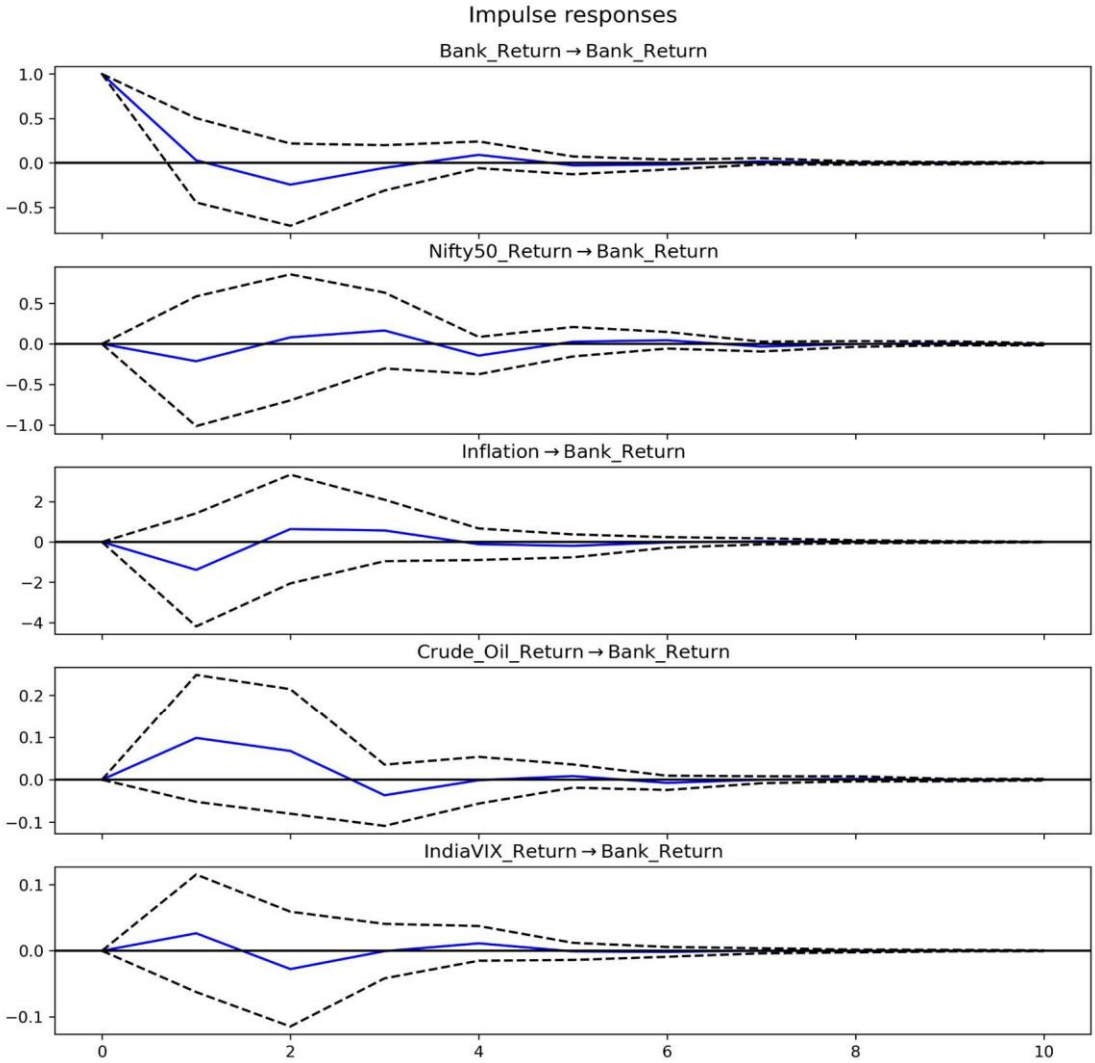
Note: IRF horizon = 10 months; table reports Months 0–5 for the most relevant shocks.

For the Banking sector, the IRF results show that an inflation shock generates a substantial negative response in Month 1 (-1.383), followed by a positive correction in Month 2 (0.641) and Month 3 (0.570), before gradually converging toward zero. This pattern supports the OLS and quantile evidence that inflation is the most relevant sector-specific macroeconomic variable for Banking.

For FMCG, the key dynamic shock is crude oil. A crude-oil shock produces a response of 0.051 in Month 1, 0.061 in Month 2, -0.014 in Month 3, and -0.031 in Month 4, before converging toward zero. This dynamic evidence complements the OLS and quantile findings that crude oil is the most economically meaningful macro shock for the FMCG sector.

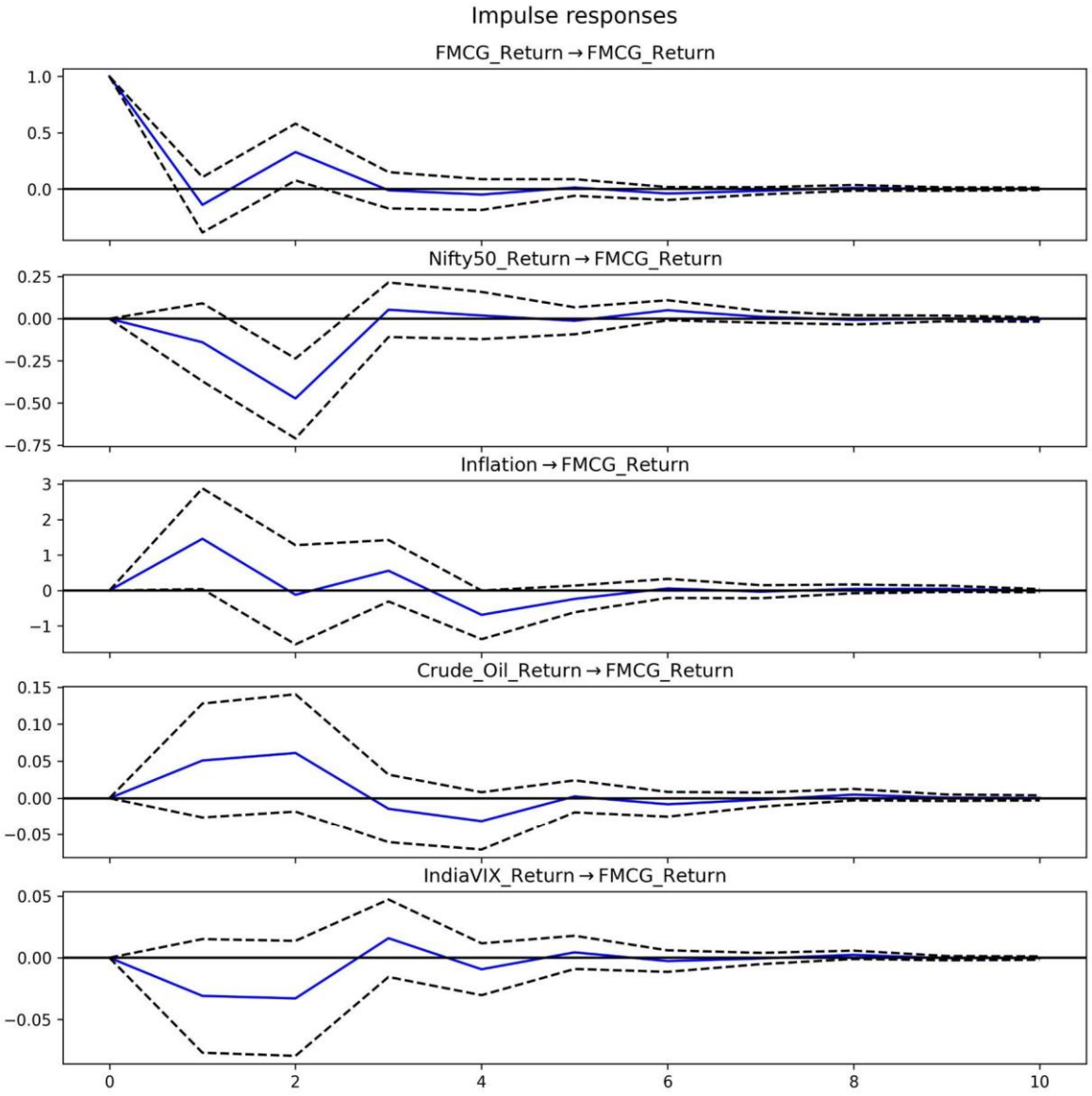
For IT and Auto, the dynamic evidence is more mixed. These sectors continue to be explained primarily through the market return channel rather than through one dominant macroeconomic variable. Across all sectors, the IRF values approach zero by the end of the horizon, suggesting that the systems are dynamically stable and that the selected macroeconomic shocks are primarily short-run in nature.

Figure 4.1. Impulse Response of Bank_Return to Inflation Shock




Source: Author's computation based on sector-wise VAR estimation.

Figure 4.2. Impulse Response of FMCG_Return to Crude Oil Shock




Source: Author's computation based on sector-wise VAR estimation.

## 4.7 Integrated Sectoral Discussion

When the results from descriptive statistics, correlation analysis, OLS regression, quantile regression, and IRF are considered together, a clear sectoral pattern emerges. The Banking sector is the strongest and most internally coherent model in the dissertation: it exhibits the highest volatility and kurtosis, the strongest correlation with Nifty50, a high market beta in OLS, a negative inflation coefficient in both OLS and quantile regression, and the clearest inflation shock response in IRF. Taken together, these results indicate that the Banking sector is both.

The FMCG sector displays a more defensive return structure in the descriptive statistics, but the econometric evidence identifies a distinct crude-oil channel. Auto is strongly market linked and moderately volatile, while IT remains the least domestically macro explained sector in the study.

## 4.8 Chapter Summary

The principal findings of this chapter can be summarized as follows: Nifty50_Return is the most consistent and statistically significant determinant of sectoral returns across all four sectors; Bank_Return is additionally and negatively affected by Inflation; FMCG Return is negatively affected by Crude Oil Return, especially in lower-return states; IT and Auto are primarily market-driven sectors; and IRF results show that sectoral responses to macroeconomic shocks are mainly short-run and converge toward equilibrium over time.

# CHAPTER 5: FINDINGS, CONCLUSION, RECOMMENDATIONS, IMPLICATIONS, LIMITATIONS, AND SCOPE FOR FUTURE RESEARCH

## 5.1 Introduction

This chapter presents the final synthesis of the study. It brings together the major findings from the empirical analysis, develops the overall conclusion, provides practical and academic recommendations, and discusses the broader implications of the results. It also acknowledges the limitations of the study and identifies directions for future research.

## 5.2 Major Findings of the Study

The descriptive analysis shows clear heterogeneity across sectors in terms of volatility and distributional characteristics. The Banking sector is the most volatile and fat-tailed, FMCG is comparatively defensive, Auto is cyclical and market-sensitive, and IT occupies a moderate position.

The OLS results indicate that Nifty50_Return is the most consistent and statistically significant determinant of sectoral stock returns across all four sectors. For the Banking sector, Nifty50_Return is positive and highly significant, while Inflation is significant and negative. For FMCG, Nifty50_Return remains significant and Crude_Oil_Return is negative and significant. For IT and Auto, the dominant explanatory variable in the baseline model is Nifty50_Return.

The wider quantile results show that macroeconomic effects are not uniform across the return distribution. Inflation is more relevant for Banking in lower and median quantiles, while crude oil is most relevant for FMCG in the lower tail. IRF analysis further confirms that the Banking sector exhibits the clearest dynamic response to inflation shocks, whereas FMCG exhibits the clearest dynamic response to crude-oil shocks. Across all sectors, responses converge toward zero over time, indicating dynamic stability.

## 5.3 Hypothesis-Wise Findings

Table 5.1

Hypothesis Test Summary

| Hypothesis | Variable | Bank | IT | FMCG | Auto | Overall Verdict |
|---|---|---|---|---|---|---|
| H1 | Exchange Rate | Not supported | Not supported | Not supported | Not supported | Not supported in any sector |
| H2 | Nifty50 Return | Supported | Supported | Supported | Supported | Strongly supported in all sectors |
| H3 | Inflation | Supported (negative) | Not supported | Not supported | Not supported | Partially supported — Bank only |
| H4 | Govt Bond 10Y | Not supported | Not supported | Not supported | Not supported | Not supported in any sector |
| H5 | Crude Oil Return | Not supported | Not supported | Supported (negative) | Not supported | Partially supported — FMCG only |
| H6 | IndiaVIX Return | Not supported | Not supported | Not supported | Not supported | Not supported at 5% level |

Source: Author's interpretation based on OLS, quantile regression, and sector-wise empirical evidence.

## 5.4 Conclusion of the Study

The dissertation concludes that broad market return is the dominant force shaping sectoral returns, but selected macroeconomic variables exert important sector-specific and statedependent effects. Banking is the sector most clearly affected by inflation, while FMCG is the sector most clearly affected by crude oil. IT and Auto remain primarily market-driven within the baseline empirical framework.

The study also demonstrates that no single econometric tool is sufficient on its own. OLS identifies average effects, quantile regression captures return-state asymmetry, and IRF reveals the time-path of dynamic transmission. Together, these methods provide a more complete understanding of how macroeconomic forces influence sectoral returns.

## 5.5 Recommendations of the Study

Investors should not treat all sectors uniformly when making portfolio decisions. Banking exposure should be monitored more carefully during inflationary periods because inflation negatively affects Bank_Return. Investors in FMCG stocks should track crude oil movements because oil-related cost shocks matter for FMCG performance. Portfolio managers should incorporate sector-

specific macro sensitivity into allocation decisions instead of assuming identical exposure across sectors.

Policymakers should note that macroeconomic conditions do not affect all sectors equally in the equity market. Inflationary conditions appear especially relevant for Banking, while commodityprice shocks are more relevant for FMCG. Sector-specific observation may therefore improve understanding of policy impact and market stability.

## 5.6 Implications of the Study

Managerially, the study shows that macro-aware and sector-specific planning is important. For financial firms, inflation and policy-sensitive conditions appear especially important. For FMCG firms, oil-related cost exposure remains a relevant strategic concern.

Theoretically, the results support the broad logic of Arbitrage Pricing Theory by showing that multiple systematic factors can matter for stock returns, but that their relevance differs by sector. Methodologically, the dissertation demonstrates the value of combining OLS, quantile regression, and IRF within one empirical framework.

## 5.7 Limitationsof the Study

The study is limited to four NSE sectoral indices and therefore does not cover the full diversity of the stock market. The baseline specification includes only a selected set of macroeconomic variables, monthly data are used instead of higher-frequency data, and the IRF analysis is conducted in relatively small sector-wise systems in order to preserve clarity and stability.

## 5.8 Scope for Future Research

Future studies may extend the sectoral coverage to include additional NSE sectors, broaden the macroeconomic set to include repo rate, GDP growth, industrial production, or foreign institutional flows, and apply higher-frequency data or more advanced econometric models such as panel quantile regression, GARCH-family models, or regime-switching frameworks.

## 5.9 Final Concluding Remark

The dissertation demonstrates that sectoral stock returns are shaped by both market-wide and macroeconomic forces, but not in a uniform way. Broad market return remains the strongest and most consistent determinant across sectors, yet inflation and crude oil emerge as distinct

sectorspecific risk factors for Banking and FMCG respectively. The principal contribution of the dissertation lies in showing that macro-financial analysis becomes more meaningful when it moves beyond the aggregate index and examines sectors individually through a multi-method empirical framework.

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# APPENDICES

Appendix A  Supplementary Excel Workbook

The accompanying Excel workbook includes price data, return data, descriptive statistics, correlation matrices, OLS outputs, quantile regression outputs, IRF values,  IRF graphs, variable definitions, methodology summary,  references and data sources, and final findings summary.

Appendix B  IRF Graphs

The IRF graphs used in Chapter 4 are reproduced in the workbook and can also be inserted into presentation slides or viva material as required.

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